

BHEL Web Development Internship Syllabus

Program Overview

This intensive one-month program is designed to equip interns with practical knowledge and hands-on experience in modern full-stack web development. By integrating industrial use cases from Bharat Heavy Electricals Limited (BHEL), participants will transition from frontend fundamentals to advanced backend services.

- **Organization:** Bharat Heavy Electricals Limited (BHEL)
- **Duration:** 1 Month
- **Primary Focus:** Full Stack WEB Development
- **Location:** 📍 Place

Curriculum Modules

Module	Topic	Key Learning Objectives
1	HTML	Understanding semantic structure (, ,) and form elements for industrial data entry.
2	CSS	Mastering selectors, box models, and layout engines like Flexbox and Grid for complex dashboards.
3	JavaScript	Implementing DOM manipulation and

Module	Topic	Key Learning Objectives
		client-side form validation for real-time interactivity.
4	Bootstrap	Utilizing the 12-column responsive grid system and pre-built components for rapid UI development.
5	Spring Boot	Developing RESTful services, handling dependency injection, and managing persistent storage with JPA.

Capstone Project: BHEL Asset Management System

As part of the final assessment, interns will develop a comprehensive industrial project modeled after BHEL's inventory dashboards.

Project Requirements

- **Frontend:** A responsive user dashboard built with React or Angular using Bootstrap for styling.
- **Backend:** A Spring Boot application providing RESTful services and business logic.
- **Database:** A MySQL schema consisting of **BHEL_Assets**, **Users**, and **Maintenance_Logs** tables.
- **Security:** Implementation of user authentication (Admin/Staff) and prevention of SQL injection.

Career Readiness & Assessment

The program concludes with a focus on professional development to prepare interns for technical roles in the industry.

- **Version Control:** Proficiency in Git workflows, including initialization, branching, and conflict resolution.
- **Mock Scenarios:** Handling industrial system failures, such as managing exceptions during sensor data crashes.
- **Interview Preparation:** Technical reviews on Java fundamentals, Spring Boot architecture, and behavioral HR strategies.